

Stop When the Evidence Is Enough

Sequential evaluation for LLMs and agents

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Aaliyan Shaikh · Ryan Marinelli

The problem

Benchmarking is expensive.

Every model × every example × every run.

`pip install bayesbench` — stop as soon as the posterior evidence is enough.

Fixed-budget evaluation

2 models × 31,800 problems (NorEval) = **63,600 calls**

$$\text{cost} = N \text{ problems} \times 2 \text{ models} \times \text{cost-per-call}$$

Most of the budget is spent **confirming a conclusion you already reached at example 40.**

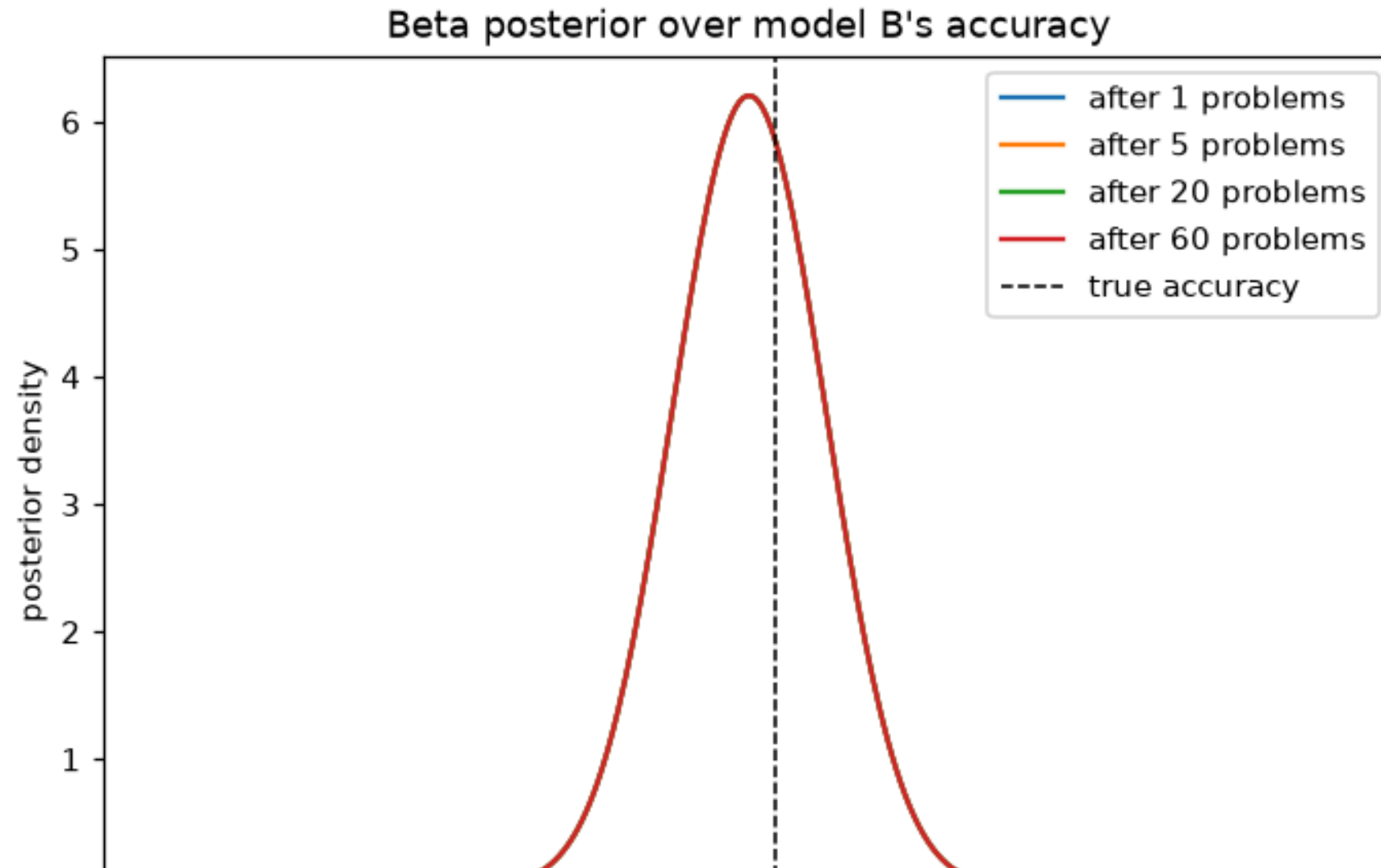
Question: can we stop earlier *and* still be right?

The idea: Bayesian sequential testing

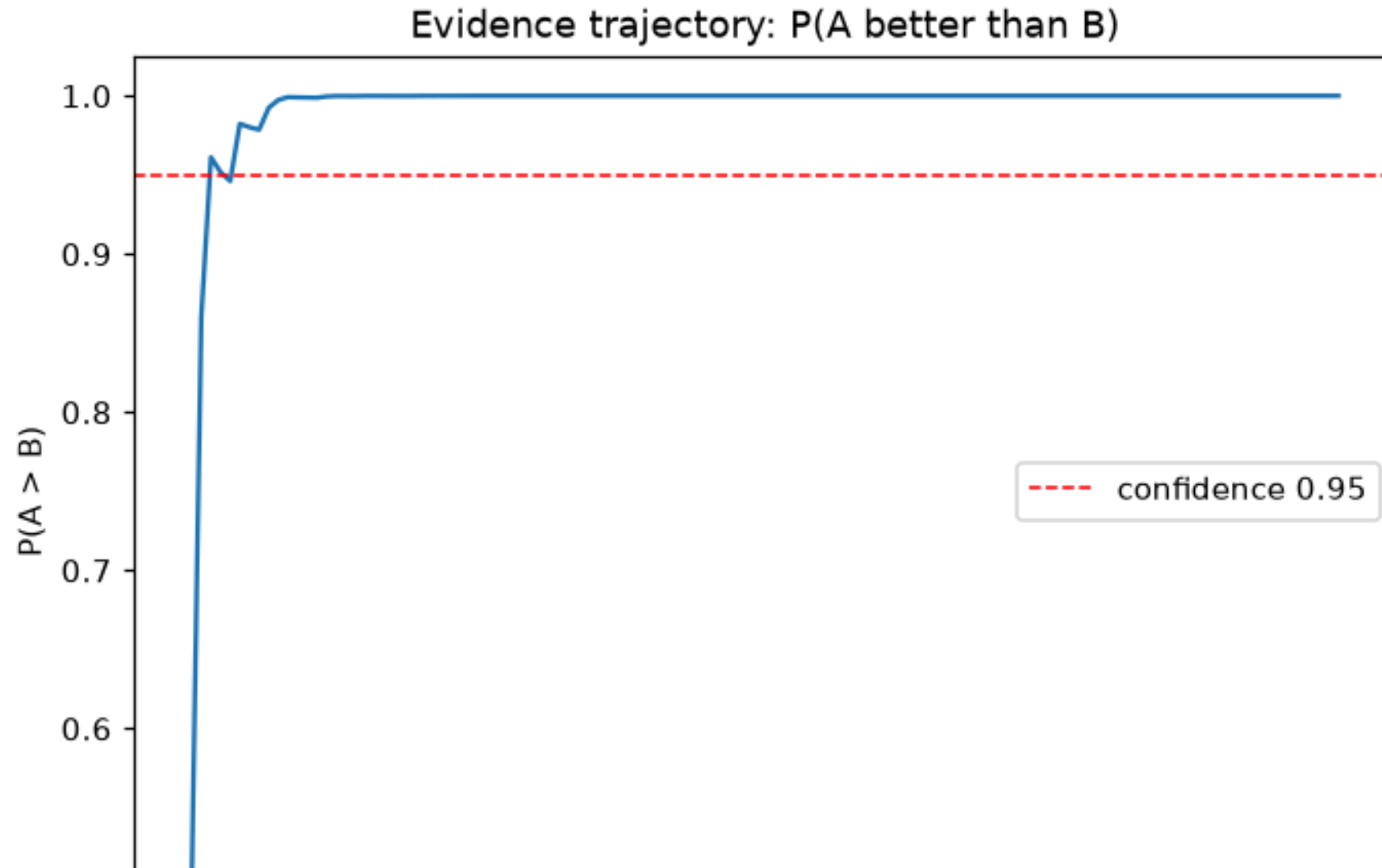
```
for each problem:  
  evaluate A and B  
  update posterior over each model's true score  
  if evidence is strong enough:  
    stop
```

- **Posterior:** Beta-Bernoulli (binary), Normal-Inverse-Gamma (continuous)
- **Evidence:** $P(A > B)$
- **Decision:** winner / equivalent / inconclusive

Posterior updating, visualized



The evidence trajectory



Three decisions, not two

Status	Meaning
winner_a / winner_b	$P(A > B)$ crossed the threshold
equivalent	posterior mass inside a ROPE margin
inconclusive	budget exhausted, not enough evidence

$P(A > B) \approx 0.5$ is **NOT** equivalence.

Early in a run, 50/50 usually means *not enough data*.

The old defaults were dangerous

`min_samples=3, skip_threshold=0.85` — every binary comparison terminated at example 3:

- 78.125% labeled "non-discriminating"
- 21.875% assigned a winner
- 0% allowed to continue

under two models that were **provably equal**.

Safe defaults: `min_samples=30`, skip heuristic off.

The optional-stopping trap

Threshold rule under **equal models** (null), skip off:

min_samples	false-winner rate
3	65.8%
30	44.1%
100	27.5%

$P(A > B)$ is a random walk: **it eventually crosses any threshold.**

Raising min_samples postpones the crossing — it doesn't remove it.

The fix: confidence sequences

`decision_rule="confidence_sequence"` (Waudby-Smith & Ramdas, 2023)

$P(\text{wrong winner at any stopping time}) \leq \alpha$ — by construction.

No calibration. No horizon assumptions. No order sensitivity.

gap	any-time false rate	E[samples]
0.00	0.0%	—
+0.20	0.0%	185
+0.40	0.0%	55

Guarantees cost samples; heuristics cost trust.

Order sensitivity is real

3 easy problems at the front of a 103-problem dataset:

- old rule: `winner_a` after **3 problems**
- true accuracy: $A = 3/103$, $B = 100/103$
- **100% wrong-winner rate**, 97% "saved"

On real NorEval data: permuting item order produces wrong winners in up to **2.3% of runs** — despite enormous true gaps.

The NorEval case study

Published: 410 / 31,800 problems, 98.7% "saving"

- Reproduction with bayesbench: **exact match**
- All 5 decisions correct — but full-corpus $P(11B > 7B) = 1.000$
- Every stop landed in the **first prompt variant** of its task
- Savings are **prospective**, never evidence of reliability

Calibration is how you know which regime you're in.

Honest evaluation workflow

1. **Large expected gap** → threshold rule + calibration report
2. **Public claim** → confidence-sequence rule ($\leq 5\%$ any-time)
3. **Shared items** → `paired=True` (per-item differences)
4. **Latency / cost** → `lower_is_better=True`
5. Always export the **decision trace** — every number reproducible

Live, traced, honest by construction

```
for update in bench.iter_compare(model_a, model_b, score_fn, dataset):  
    print(update.trace.step, update.trace.p_a_beats_b)
```

- Models are called **lazily** — reported samples == actual calls
- Every step: scores, $P(A > B)$, status, terminal reason
- Export: JSON / CSV traces

What you'll do in the notebook

1. Fixed-budget evaluation and its cost
2. Posterior updating, visualized
3. Superiority vs equivalence vs uncertainty
4. Calibration across seeds and effect sizes
5. Live CPU-only comparison with traces
6. Exercises: confidence, margins, ordering, cost/reliability

`stop_when_the_evidence_is_enough.ipynb` + `exercises/`

Key takeaways

- Stop early when the gap is large; refuse to decide when it isn't.
- $P(A > B) \approx 0.5$ is insufficient evidence — equivalence needs a ROPE.
- First-crossing rules can be gamed by order and fooled by noise.
- Confidence sequences give $\leq 5\%$ false decisions at any stopping time.
- Report savings as prospective, next to your calibration numbers.

References

- tinyBenchmarks — *Maia Polo et al.*, ICML 2024
- MixEval — *Ni et al.*, NeurIPS 2024
- How Benchmark Prediction from Fewer Data Misses the Mark — NeurIPS 2025
- Waudby-Smith & Ramdas, *Estimating means of bounded random variables by betting*, JRSS-B 2023
- Ville (1939) — the martingale foundation

bayesbench: <https://github.com/rymarinelli/bayesbench>

Thank you

Questions → bring your benchmarks.

```
pip install bayesbench
```